Amirali Rayegan

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Education

North Carolina State University Jan. 2024 – Jan. 2028 (Expected)
Ph.D. in Computer Science, Advisor: Dr. Tim Menzies, GPA: 4.00/4.00 *Raleigh, NC*

- Graduate Merit Award recipient, NC State College of Engineering (2024, 2025).

North Carolina State University Jan. 2024 – May 2026
M.S. in Computer Science, GPA: 4.00/4.00 *Raleigh, NC*

University of Tehran Sep. 2018 – May 2023
B.S. in Computer Engineering, Information Technology, GPA: 3.62/4.00 *Tehran, Iran*

Research Experience

North Carolina State University Jan. 2024 – Present
Graduate Research Assistant *Raleigh, NC*

- **Stability and Causality in Software Analytics:** Extended EZR to study model instability. Identified settings that increased agreement by **4.8**× and reduced optimization-error variability by **22%**, while investigating causal interventions for the remaining instability.
- **Trajectory-Assisted Code Review for Agentic AI:** Developing an LLM-powered code review research pipeline that uses agentic AI execution traces, automated feature extraction, and statistical analysis to evaluate trajectory-assisted review of AI-generated code.
- **Label-Efficient Multi-Objective Optimization:** Introduced **EZR**, a label-efficient and interpretable optimization method that matched or exceeded state-of-the-art approaches while reducing labeling requirements by **91%** on average, reducing feature spaces by **90%**, and retaining approximately **80%** of referenced optimum performance.
- **Explanations for Search-Based Software Engineering:** Building on the optimization and stability studies, conducting ongoing research on explanations for SBSE, what information explanations should prioritize, which factors matter most, and how explanation quality can be quantitatively evaluated.
- **Multi-Objective Optimization Infrastructure:** Co-developed **MOOT**, a repository of **127 real-world multi-objective optimization tasks** for evaluating LLM systems and AI agents, spanning software configuration, cloud systems, project health, and process models.

University of Tehran May 2021 – Dec. 2022
Undergraduate Research Assistant *Tehran, Iran*

- **Financial Forecasting:** Comparative study of ML models on U.S. stock-price prediction (including ARIMA, SVM, RNN, LSTM, and Bi-LSTM)
- **Ethereum Transition:** Investigated the Ethereum's blockchain transition from proof-of-work to proof-of-stake.

Publications

- **A. Rayegan**, L. Li, T. Menzies. Do Causal-Inspired Optimizers Improve Optimization Quality and Consistency? *Under review, SANER 2027.*
- **A. Rayegan**, T. Menzies. Can Causality Cure Confusion Caused by Correlation? *Registered report*
- **A. Rayegan**, L. Li, T. Menzies. Is Model Instability just Noise to be Tolerated or a Property that can be Managed? *Under review, ICSE 2027.*
- **A. Rayegan**, T. Menzies. Minimal data, Maximum Clarity: A Heuristic for Explaining Optimization *Published in Journal of Systems and Software (JSS 2026).*
- T. Menzies, T. Chen, Y. Ye, K. K. Ganguly, **A. Rayegan**, S. Srinivasan, A. Lustosa. MOOT: a Repository of Many Multi-Objective Optimization Tasks *Published in Mining Software Repositories (MSR2026).*
- **A. Rayegan**, A. Shiri, B. Bahrak.A Comparative Study of ML Techniques for Stock Price Prediction *ICSPIS 2022*

Skills

Programming & Tools: Python, C++, SQL, Git, Linux, PyTorch, TensorFlow, scikit-learn, NumPy, pandas
Research: AI for Software Engineering, Applied Machine Learning, Multi-objective optimization, causal methods, Explainable AI(XAI), statistical modeling, Trajectory analysis, Agentic AI workflow, LLM Evaluation